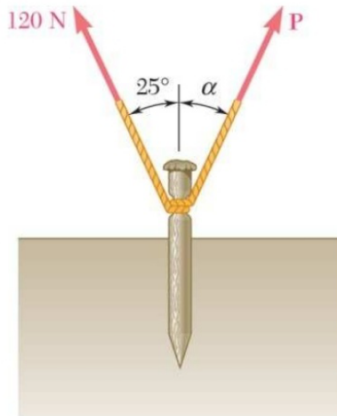


Problem 2.5: A stake is being pulled out of the ground by means of two ropes as shown. Knowing that $\alpha = 30^\circ$, determine by trigonometry (a) the magnitude of the force P so that the resultant force exerted on the stake is vertical, (b) the corresponding magnitude of the resultant.



힘력이 수직이 되도록 작용한다고 가정하고 그림을 그리면



Sine law을 사용

$$\frac{120}{\sin 30^\circ} = \frac{P}{\sin 25^\circ} = \frac{Q}{\sin 125^\circ}$$

$$a) \therefore P = 120 \times \frac{1}{\sin 30^\circ} \times \sin 25^\circ = 101.4 \text{ N}$$

$$b) Q = 120 \times \frac{1}{\sin 30^\circ} \times \sin 125^\circ = 196.6 \text{ N}$$

Problem 2.10: Two forces are applied as shown to a hook support. Knowing that the magnitude of P is 35 N, determine by trigonometry (a) the required angle α if the resultant R of the two forces applied to the support is to be horizontal, (b) the corresponding magnitude of R .

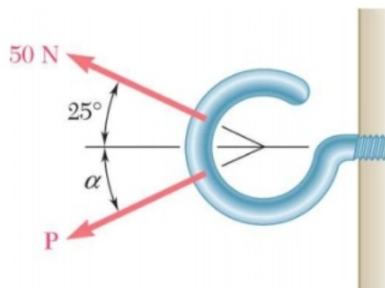
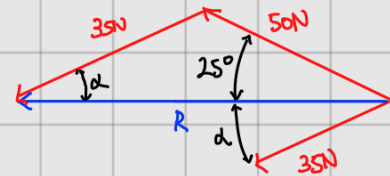


Fig. P2.10

힘력이 수평이 되도록 작용하도록 그림을 그리면



Sine law을 사용

$$\frac{35}{\sin 25^\circ} = \frac{50}{\sin \alpha} = \frac{R}{\sin (155^\circ - \alpha)}$$

$$\therefore \sin \alpha = 50 \times \frac{1}{35} \times \sin 25^\circ$$

$$= 0.6$$

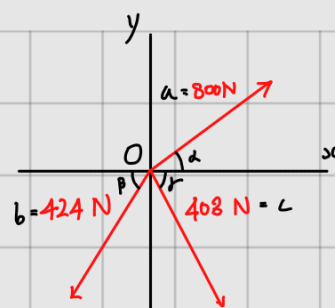
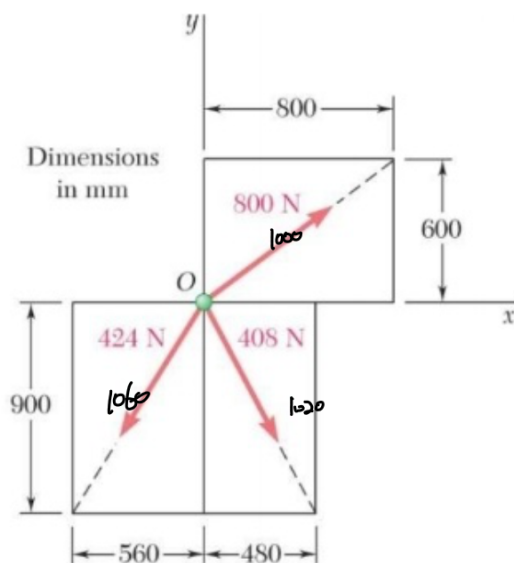
$$a) \therefore \alpha = \sin^{-1} 0.6 = 36.9^\circ$$

$$R = \sin (155^\circ - \alpha) \times \frac{1}{\sin 25^\circ} \times 35$$

$$= \sin (118.1^\circ) \times \frac{1}{\sin 25^\circ} \times 35$$

$$b) R = 73.1 \text{ N}$$

Problem 2.22: Determine the x and y components of each of the forces shown.



이러한 식변

$$\sin \alpha = \frac{3}{5} \quad \cos \alpha = \frac{4}{5}$$

$$\sin \beta = \frac{45}{53} \quad \cos \beta = \frac{28}{53}$$

$$\sin \gamma = \frac{15}{17} \quad \cos \gamma = \frac{8}{17}$$

$$\therefore a_x = a \cos \alpha = 800 \times \frac{4}{5} = 640 \text{ N}$$

$$\vec{a}_x = 640 \hat{i}$$

$$a_y = a \sin \alpha = 800 \times \frac{3}{5} = 480 \text{ N}$$

$$\vec{a}_y = 480 \hat{j}$$

$$b_x = b \cos \beta = 424 \times \frac{28}{53} = 224 \text{ N}$$

$$\vec{b}_x = -224 \hat{i}$$

$$b_y = b \sin \beta = 424 \times \frac{45}{53} = 360 \text{ N}$$

$$\vec{b}_y = -360 \hat{j}$$

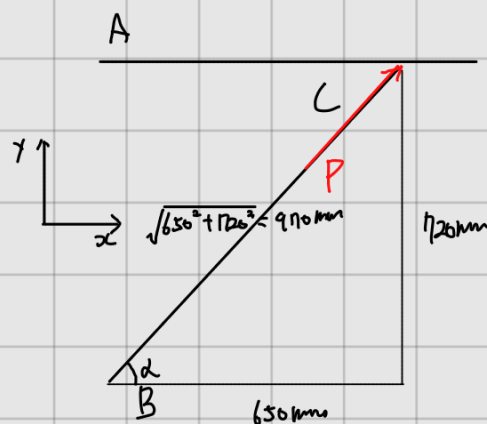
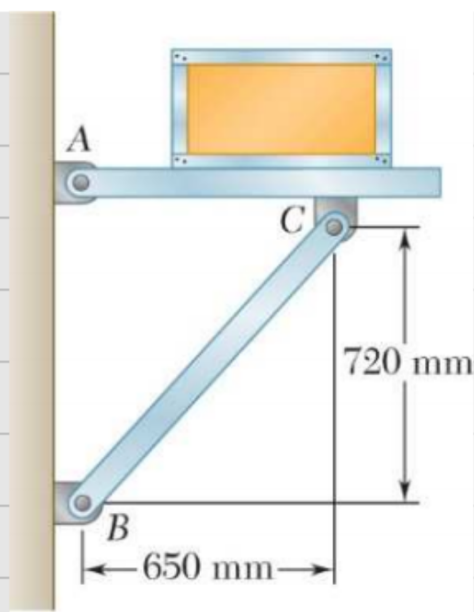
$$c_x = c \cos \gamma = 408 \times \frac{8}{17} = 192 \text{ N}$$

$$\vec{c}_x = 192 \hat{i}$$

$$c_y = c \sin \gamma = 408 \times \frac{15}{17} = 360 \text{ N}$$

$$\vec{c}_y = -360 \hat{j}$$

Problem 2.25: Member BC exerts on member AC a force $\mathbf{P}$ directed along line BC . Knowing that $\mathbf{P}$ must have a 325-N horizontal component, determine (a) the magnitude of the force $\mathbf{P}$, (b) its vertical component.



$$P_x = 325 \text{ N}$$

$$\sin \alpha = \frac{72}{97}$$

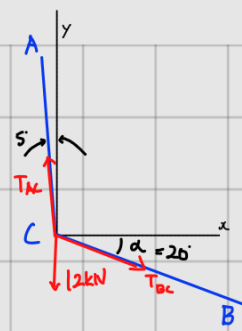
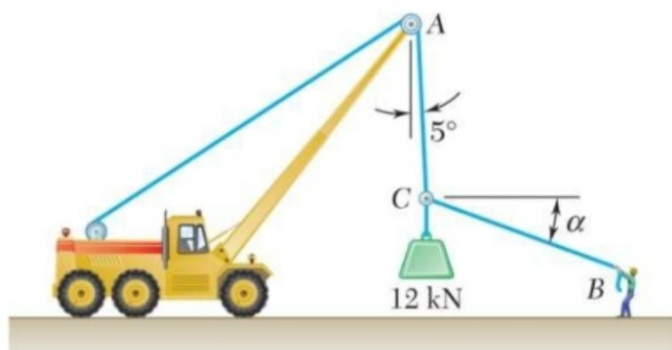
$$\cos \alpha = \frac{65}{97}$$

$$P_x = P \cos \alpha = 325$$

$$a) \therefore P = \frac{325 \text{ N}}{\cos \alpha} = 325 \times \frac{97}{65} = 485 \text{ N}$$

$$b) P_y = P \sin \alpha = 485 \text{ N} \times \frac{72}{97} = 360 \text{ N}$$

Problem 2.48: Knowing that $\alpha = 20^\circ$, determine the tension (a) in cable AC, (b) in rope BC.



point C가 정지해 있으니까 Force Vector는 삼각형을 이룬다.

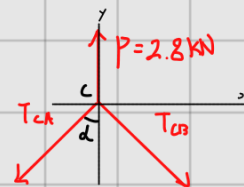
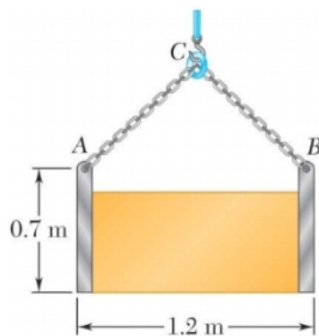
Sine law이랑

$$\frac{12 \text{ kN}}{\sin 115^\circ} = \frac{T_{AC}}{\sin 110^\circ} = \frac{T_{BC}}{\sin 5^\circ}$$

$$T_{AC} = \frac{12 \text{ kN}}{\sin 115^\circ} \times \sin 110^\circ = 11.7 \text{ kN}$$

$$T_{BC} = \frac{12 \text{ kN}}{\sin 115^\circ} \times \sin 5^\circ = 1.08 \text{ kN}$$

Problem 2.61: A movable bin and its contents have a combined weight of 2.8 kN. Determine the shortest chain sling ABC that can be used to lift the loaded bin if the tension in the chain is not to exceed 5 kN.



alpha가 클 수록 chain이 짧아짐.

$$T = T_{CA} = T_{CB} \leq 5 \text{ kN}$$

C점이 잠겨있을 때 이므로

$$\sum (F_x)_y = 0$$

$$T_y = (T_{CA})_y = (T_{CB})_y = T_{CA} \cos \alpha = T_{CB} \cos \alpha = T \cos \alpha$$

$$\sum (F_c)_y = -(T_{CA})_y - (T_{CB})_y + P = 0$$

$$-2T_y + P = 0$$

$$\therefore T_y = 1.4 \text{ kN}$$

$$\therefore T = \frac{1.4 \text{ kN}}{\cos \alpha} \leq 5 \text{ kN}$$

$$\therefore \frac{1.4}{5} \leq \cos \alpha$$

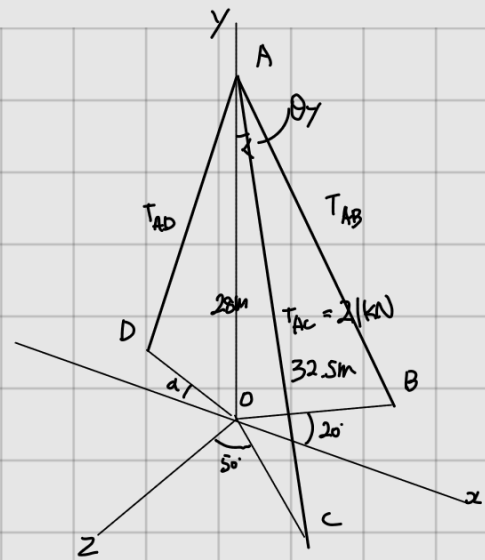
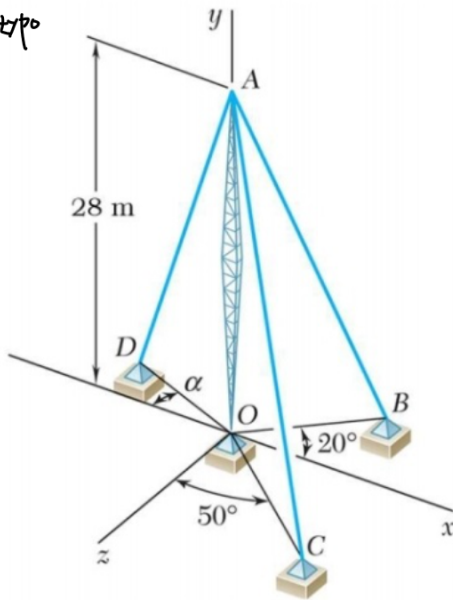
$$\alpha \leq \cos^{-1} \left(\frac{1.4}{5} \right)$$

$$\alpha \leq 73.7^\circ$$

$$\therefore \alpha = 73.7^\circ$$

Problem 2.78: Cable ~~AB~~^{AC} is 32.5 m long and the tension in that cable is 21 kN. Determine (a) the x , y , and z components of the force exerted by the cable on the anchor C , (b) the angles θ_x , θ_y and θ_z , defining the direction of that force.

There is a typo



$$\cos \theta_y = \frac{28}{32.5}$$

$$b)-1 \therefore \theta_y = \cos^{-1}\left(\frac{28}{32.5}\right) = 30.51^\circ$$

Projection onto y -axis

$$\begin{aligned} (T_{AC})_y &= T_{AC} \cos \theta_y \\ a)-1 &= 21 \text{ kN} \times \frac{28}{32.5} = 18.1 \text{ kN} \end{aligned}$$

$\therefore$ Projection onto z - x plane

$$\begin{aligned} (T_{AC})_{z-x} &= T_{AC} \sin \theta_y \\ &= 21 \text{ kN} \times \sin(30.51^\circ) \\ &= 10.66 \text{ kN} \end{aligned}$$

$$\begin{aligned} \therefore (T_{AC})_x &= -(T_{AC})_{z-x} \sin 50^\circ \\ &= -10.66 \sin 50^\circ \end{aligned}$$

$$a)-2 = -8.17 \text{ kN}$$

$$\begin{aligned} \therefore (T_{AC})_z &= -(T_{AC})_{z-x} \cos 50^\circ \\ &= -10.66 \cos 50^\circ \end{aligned}$$

$$a)-3 = -8.65 \text{ kN}$$

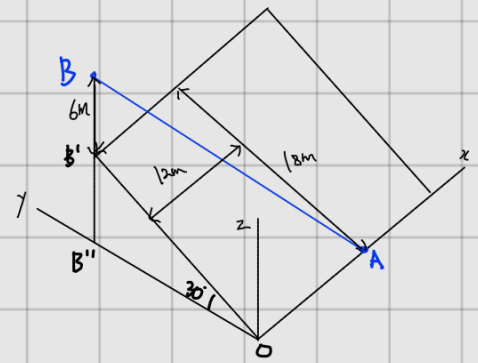
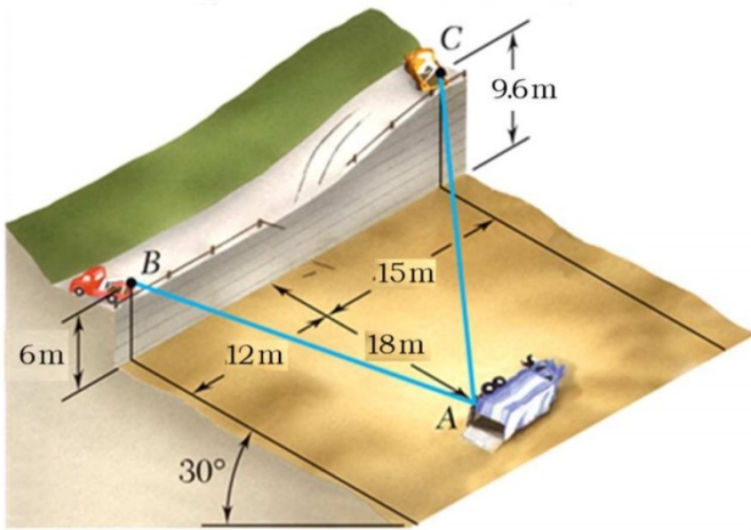
$$\cos \theta_z = \frac{(T_{AC})_z}{T_{AC}} = \frac{-8.17}{21}$$

$$b)-2 \therefore \theta_z = 112.9^\circ$$

$$\cos \theta_z = \frac{(T_{AC})_z}{T_{AC}} = \frac{-8.65}{21}$$

$$b)-3 \therefore \theta_z = 114.3^\circ$$

Problem 2.87: In order to move a wrecked truck, two cables are attached at A and pulled by winches B and C as shown. Knowing that the tension in cable AB is 10 kN, determine the components of the force exerted at A by the cable.



$$\overline{OB'} = 12\text{m}$$

$$\overline{B'A} = 18 \sin 30^\circ = 9\text{m}$$

$$\overline{OA} = 18 \cos 30^\circ = 9\sqrt{3}$$

$$\vec{OA} = 12\hat{i} + 15\hat{k}$$

$$\vec{OB} = 0\hat{i} + 9\sqrt{3}\hat{j} + 15\hat{k}$$

$$\vec{AB} = -\vec{OA} + \vec{OB} = -12\hat{i} + 9\sqrt{3}\hat{j} + 15\hat{k}$$

$$\overline{AB} = \sqrt{(-12)^2 + (9\sqrt{3})^2 + (15)^2} = 24.74$$

$$\therefore \lambda_{AB} = \frac{\vec{AB}}{\overline{AB}} = \frac{1}{24.74} (-12\hat{i} + 9\sqrt{3}\hat{j} + 15\hat{k})$$

$$\vec{T}_{AB} = T_{AB} \lambda_{AB}$$

$$= 10\text{ kN} \cdot \frac{1}{24.74} (-12\hat{i} + 9\sqrt{3}\hat{j} + 15\hat{k})$$

$$\text{ans.} = -4.85\hat{i} + 6.30\hat{j} + 6.06\hat{k}$$